Yulong Liu

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Research Interests

AI methods & discovery: Differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, and AI for Science and scientific discovery.

Engineering: Homogenization and multiscale modeling, coupled multiphysics simulation, and subsurface energy systems, including geothermal reservoirs and hydraulic fracturing.

Education

Ph.D. in Earth Science – Cornell University 2024 – Present
Minors: Computer Science and Scientific Computing
B.S. in Mining Engineering – Northeastern University (CN) 2024

Research Experience

Ph.D. Candidate – Dept. of Earth and Atmospheric Sciences, Cornell University Aug. 2024 – Present
Advisor: Chloé Arson *Ithaca, NY*

- Develop coupled thermal, hydraulic and mechanical models in MOOSE for geothermal reservoirs.
- Build joint temperature–pressure neural operators for geothermal forecasting, uncertainty propagation, Bayesian reservoir inference and monitoring design.
- Study differentiable programming and PDE neural solvers, integrating physics-informed networks, implicit neural representations and neural operators with numerical simulation.
- Investigate parameterization of theoretical homogenization and multiscale transport in porous media, using scientific machine learning to discover interpretable effective permeability laws.
- Advance agentic LLMs for science; lead a cross-institution project on physics-grounded verification that generated multiphysics simulation code represents the intended physics.

Undergraduate Researcher – Northeastern University (CN) Sep. 2020 – Jun. 2024
B.S. in Mining Engineering *Shenyang, China*
Thesis: A Joint Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning

- Developed a joint deep-learning model for microseismic signal classification and arrival-time picking, integrating tasks typically handled in separate workflows.
- Conducted experimental and computational research on hydraulic fracturing, triaxial rock testing, and tunnel boring machines; received the Outstanding Thesis Award and National Scholarship.

Research Projects and Funding

Joint Operator Learning and Digital Twins for Enhanced Geothermal Systems 2025 – Present
Coupled reservoir simulations, joint temperature–pressure forecasting and observation-based permeability inference for the Cornell geothermal case study.

- **Cornell Atkinson Academic Venture Fund:** *AI-aided Drilling for Deep Geothermal Development: Cornell Site Case Study*. PI: C. Arson; Co-PI: Y. Yang. 2025–2027.
- **NSF R2I2, Award #2517787:** *Enhancing Northeast Resilience through Innovations for Deep Geothermal Heat Direct Use*. PI: C. Arson; Co-PIs: S. Olmstead and J. Tester. 2025–2027.
- **Cornell Engineering SPROUT-energy:** *Mathematical Modeling Framework for Enhanced Geothermal Systems Digital Twins*. 2026.
- **NSF ACCESS project (EES260150):** *Learned Uncertainty-Propagation and World Models for Enhanced Geothermal Reservoirs*. PIs: **Y. Liu** and C. Arson. 1.5M ACCESS Credits (equivalent to $37.5K). 2026–2027.
- **Empire AI Alpha+Beta:** *AI-aided Computational Geomechanics*. PIs: Y. C. Han, **Y. Liu**, A. Tristani, and C. Arson. 5.5K GPU hours (equivalent to $2.75K). 2026.
- **Cornell Atkinson Graduate Research Grant** (under review): *An Action-Conditioned Geothermal Digital Twin for Adaptive Heat Extraction at Cornell*. Applicant: **Y. Liu**; faculty mentor: C. Arson. $8K requested. Proposed 2027–2028.

Theoretical Homogenization of Multiscale Porous Media Flow by Machine Learning and Foundation Models 2025 – Present

Homogenization theory and scientific machine learning for interpretable permeability laws.

- **Cornell Engineering SPROUT:** *Optimizing Metrics for Transport Homogenization: Subsurface Energy Systems Predictive Models.* Aug. 2025 – Jul. 2026.
- **NSF ACCESS project (EES260208):** *Homogenization and Machine Learning for Multiscale Porous Media Flow.* PIs: **Y. Liu** and C. Arson. 1.5M ACCESS Credits (equivalent to $37.5K). 2026–2027.
- **Simons Foundation, Utilizing the Full Power of Empire AI** (proposal in preparation; proposed 2027–2029). PI: Y. Yang; Co-PI: C. Arson. **Role:** Contributing Author and Scientific Co-Lead.
  - Invited by the faculty PI to serve as Co-PI; formal Co-PI designation was precluded by institutional and sponsor eligibility requirements for graduate students.
  - Led the scientific conceptualization, proposal development, and research design.
  - Responsible for leading the proposed research program on physics-informed foundation world models and computational multiphysics.

Groundwater Digital Twin Proposed

- **NASA ROSES-25 F.5 FINESST** (under review). Future Investigator: **Y. Liu**; PI: C. Arson. Full title withheld.

Publications

Selected Papers

- **Y. Liu**, J. Botvinick-Greenhouse, Y. Yang, C. Arson. *Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection–Production Systems: A Cornell Case Study.* ARMA US Rock Mechanics/Geomechanics Symposium, Tucson, AZ, 2026.
- **Y. Liu**, C. Arson. *A Physics-Informed Neural Network for Pressurized Cavities of Arbitrary Shape in Heterogeneous Rock. Rock Mechanics and Rock Engineering* (2026). doi:10.1007/s00603-026-05882-5.
- M. Belachew, **Y. Liu**, J. D. Frost, C. Arson. *Numerical Assessment of Plasticity Development and Energy Expenditure of Ant-Like Microtunnelling. Tunnelling and Underground Space Technology*, 172, 107501, 2026.
- **Y. Liu**†, C. Arson. *Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity in Homogeneous and Bilayered Media.* ARMA US Rock Mechanics/Geomechanics Symposium, D022S018R006, 2025.
- **Y. Liu***,†, Z. Song*, et al. *Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code.* arXiv:2605.09360. Under review at NeurIPS 2026.
- Z. Song, Y. Li, **Y. Liu**†. *Prefix-Safe Bayesian Belief Tracking for LLM Reasoning Reliability: Separating Calibration from Ranking.* arXiv:2605.27712. Under review at ACL ARR 2026.

Thesis

- **Y. Liu**. *A Joint Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning.* Undergraduate thesis, Northeastern University, 2024.

* Co-first author † Corresponding author

Conferences and Presentations

Invited participant, CHiMaD Phase Field Methods XXI Upcoming
Center for Hierarchical Materials Design (CHiMaD)

Participant, First Empire AI Symposium Dec. 2026
Flatiron Institute, New York, NY, USA. Abstract: "Scaling Subsurface Multiphysics Learning on Empire AI: From Neural Operators to a Physics Foundation Model"

Poster presentation, 60th US Rock Mechanics/Geomechanics Symposium (ARMA) Jun. 2026
American Rock Mechanics Association, Tucson, AZ, USA

Panelist, CHiMaD Phase Field Methods XX May 2026
Center for Hierarchical Materials Design, Northwestern University, Evanston, IL, USA

Poster presentation, Cornell CEE Graduate Research Symposium Apr. 2026
Cornell University, Ithaca, NY, USA

Invited seminar: "Physics-Informed AI Optimization of Heterogeneous Rock Systems" Dec. 2025
Southwest Petroleum University seminar series, Chengdu, China

Poster presentation, 59th US Rock Mechanics/Geomechanics Symposium (ARMA) Jun. 2025
American Rock Mechanics Association, Santa Fe, NM, USA

Fellowships and Awards

Estwing Hammer Award – Most Outstanding EAS Graduate Student of 2024, Cornell University 2025
Cornell University Travel Grant, Cornell University 2025, 2026
Outstanding Thesis, Northeastern University 2024
National Scholarship, Ministry of Education, China 2024
First Class Scholarship, Northeastern University 2023
Autumn Scholarship, Northeastern University 2022
Golden Seed Scholarship, Northeastern University 2022

Academic Service

- Session Chair, GS03 Computational Geomechanics, Coupled Thermo-Hydro-Mechanical-Chemical (THMC) Processes, and Next-Generation Modeling, Geothermal Track, *61st U.S. Rock Mechanics/Geomechanics Symposium (ARMA 2027)*, Austin, TX, June 2027
- Reviewer, *Rock Mechanics and Rock Engineering*
- Reviewer, *Underground Space*
- Reviewer, *U.S. Rock Mechanics/Geomechanics Symposium*, American Rock Mechanics Association (ARMA)
- Reviewer, *International Conference on Artificial Neural Networks*
- Reviewer, *Geothermal Rising Conference*

Skills

Programming & Scientific Computing: Python, C++, R, MOOSE finite element framework

Modeling & Simulation: finite element modeling, coupled multiphysics simulation, poromechanics, computational geomechanics, CAD (e.g., AutoCAD, SolidWorks)

Subsurface & Reservoir: geothermal reservoirs, hydraulic fracturing, injection–production modeling, homogenization and multiscale transport in porous media

Scientific Machine Learning: differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, AI for Science and scientific discovery; physics-informed networks, neural operators, and implicit neural representations

LLMs for Simulation: using large language models to accelerate simulation-software workflows and to verify the physical correctness of generated simulation code

Experimental Methods: rock mechanics testing (e.g., triaxial compression), laboratory instrumentation

Currently Learning: phase-field modeling (fracture/multiphysics), multiphase flow in porous media

Scientific Writing: LaTeX / Overleaf

Professional Memberships

- Member, American Rock Mechanics Association (ARMA), 2025–Present
- Member, International Society for Rock Mechanics and Rock Engineering (ISRM), 2025–Present
- Member, Chinese Society for Rock Mechanics and Engineering (CSRME), 2024–Present
- Member, Geothermal Rising, 2025–Present